2a	Total volume of molecules is negligible compared with volume occupied by the gas	B1
2b	$pV = NkT$ $2.10 \times 10^5 \times 950 \times 10^{-6} = N \times 1.38 \times 10^{-23} \times (280)$ $N = 5.16 \times 10^{22}$ volume of one molecule = $(4 / 3)\pi r^3$ (= 1.41×10^{-29} m^3) volume of all molecules = $5.16 \times 10^{22} \times 1.41 \times 10^{-29}$ = 7.3×10^{-7} m^3 = 7×10^{-7} m^3 (**1 s.f**.; as this is an estimation question the uncertainty of final answer cannot be more precise than that of the given data.) **OR** volume of one molecule = d^3 (= 2.7×10^{-29} m^3) volume of all molecules = $2.7 \times 10^{-29} \times 5.16 \times 10^{22}$ = 7×10^{-7} m^3 (**1 s.f**.; as this is an estimation question the uncertainty of final answer cannot be more precise than that of the given data.)	 C1 C1 A1 (C1) (C1) (A1)
2c	Since volume of all atoms (1×10^{-6} m^3) is *3 orders of magnitude less* than volume occupied by the gas (950×10^{-6} m^3 $\approx 1 \times 10^{-3}$ m^3) OR 0.1% of volume occupied by the gas so assumption in (a) is justified.	B1
2d(i)	Internal energy depends on temperature. Since final temperature equal initial temperature so **no change in total internal energy**	B1 B1
2d(ii)	<u>For P → Q:</u> work done on gas = **0 J** and increase in internal energy, $\Delta U = Q + W = +97.0 + 0 =$ **97.0 J** <u>For Q → R</u>: increase in internal energy, $\Delta U = 0 - 42.5 =$ **–42.5 J** <u>For R→P:</u> work done on gas, $W = p \Delta V = 2.10 \times 10^5 \times (1125 – 950) \times 10^{-6} =$ **36.8 J** since total change in internal energy is zero, $+97 + (–42.5) + \Delta U_{RP} = 0$ increase in internal energy, $\Delta U_{RP} =$ **–54.5 J** thermal energy supplied, $Q = \Delta U – W = −54.5 – 36.8 =$ **–91.3 J**	 A1 A1 A1 A1 A1
	Total:	12

3a	upthrust and weight	B1